The Relationship between Educational Attainment and Labor Force Participation in Switzerland

I. Introduction: introduce topic of investigation

Labor force participation rate is a measurement of the employment situation of proper labor based on the entire labor force population of a specific country. Investigating factors behind labor force participation will provide us broader understanding for employment patterns and individual's well-being. Many factors can influence labor force participation. For example, having better educational attainment will intuitively improve people's chance of joining labor market. This investigation will explore the relationship between years of formal education and labor force participation for Swiss labor through R coding.

By using R to create regressions, scatterplots, and data of balance table, the investigation will also consider how nonlabor income will influence the relationship. Data in this project comes from Swiss Labor Market Participation Data from AER dataset in R, originating from the health survey SOMIPOPS for Switzerland in 1981.

To understand the relationship between educational attainment and labor force participation in Switzerland, we need to conduct detailed regressions to analysis data. The investigation began with a basic model examining the direct relationship between education and participation, then expanded to account for various demographic and economic factors that might influence this relationship.

The findings in the investigation suggest that the decision of labor force participation involves complex factors to consider. From the available data, in Switzerland, family structure (particularly young children) and economic factors (income) are more important determinants of participation than education, indicating individual's participation in labor market involves overall household conditions.

II. Data

A. Data introduction

Observation definition is years of formal education of a Swiss individual. Explanatory Variable is education level which is measuered by years of formal education. Response Variable is labor force participation. In this investigation, labor force participation will be measured by a dummy variable which "0" represents no participation and "1" represents participation. Potential confounding variable is the age of the measured individual, nonlabor income, number of kids possessed (both aged below 7 and above 7). Treatment is years of formal education where different years represent different groups of treatment. Therefore, this investigation will explore how received different years of education will impact the likeliness of individual Swiss labor force participation.

B. Ceteris Paribus

The "Road Not Taken" refers to the counterfactual outcome that we can never directly observe. In context, the hypothetical scenario is labor force participation will be impacted on a different number of years of formal educaiton of individual Swiss, holding other charateristics constant. Consider a 30-year-old Swiss woman has 12 years of education and has two young children who is not participating in the labor force. The "road not taken" would be her hypothetical labor force participation if she had 16 years of education, holding all other characteristics constant.

There are several challenges in estimation, including selection bias among participants due to non-random education choices, and unobserved factors affecting both education and participation. Additionally, labor force participation can be a household-level decision rather than an individual decision. However, we can possibly address the challenges by adding potential confounding variables as controls into the regression line.

C. Check for balance

To check whether covariates outside the investigation may confound the relationship between education and labor force participation, we need to check the balance of potential confounding variables.

Data can be divided to two groups: one group representing individual Swiss who didn't participate in labor market, the other group representing individuals who did participate. Variables beside year of education and labor force participation include non-labor income measured in logarithm, age in decades, number of young kids aged under 7, and number of old kids aged above 7. By calculating and collecting mean and standard deviations for each variables in two seperate groups, we will use standardized mean difference (SMD) to assess whether the potential confounding variables between the two groups are balanced.

	no participation	yes participation	SDM
education	mean=9.594;sd=2.851	mean=8.97;sd=3.211	0.206
income(in log)	mean=10.751;sd=0.436	mean=10.608;sd=0.369	0.354
age(in decades)	mean=4.085;sd=1.161	mean=3.89;sd=0.905	0.188
youngkids	mean=0.41;sd=0.678	mean=0.197;sd=0.504	0.356
oldkids	mean=0.902;sd=1.063	mean=1.077;sd=0.0187	0.161

In SMD, $|SMD| < 0.1$ represents well balanced, $0.1 \leqslant |SMD| < 0.2$ represents moderate imbalance, and $|SMD| \geqslant 0.2$ represents substantial imbalance. While age and number of old kids are moderate imbalance, both non-labor income and number of young kids show substantial imbalance. While none of these variables are balanced, regression models need to add these variables as controls.

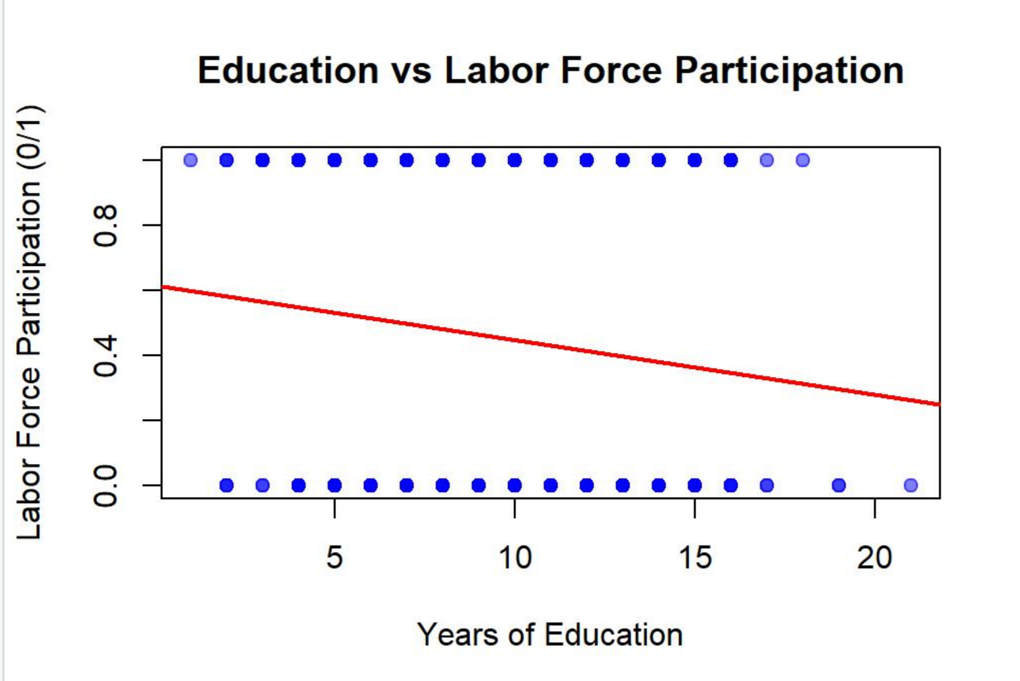
The imbalances for the covariates with participation and no participation might indicate how the covariates can influence people's decision to participate. Individuals already possessed higher income outside of participating labor market probably don't not need to participate in labor market for income; different age stages might impact deciding participation are probably due to inversed U-shape labor market participation pattern; possessing either young kids or old kids probably will affect people's choice of concentration allocation, influencing decision of participation. Therefore, there may be observed differences in the samples.

III. Regression Analysis

A. Regression models

To establish a baseline understanding that can make comparisons for more analysis, we need to construct a basic regression model that examined only the relationship between years of education and labor force participation. Basic model:

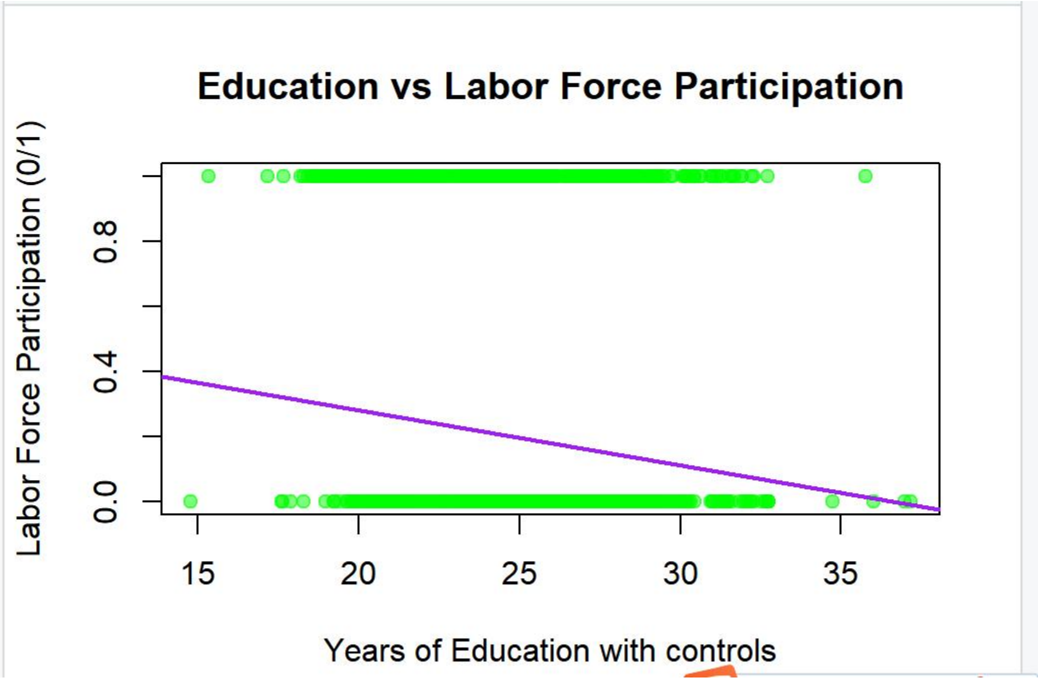
$$participation\_num = \beta_0 + \beta_1 (education) + \varepsilon$$



β_0 represents the predicted labor force participation when education years is zero, which is a baseline measurement. β_1 is coefficient of education, meaning on how much will the level of education impact individual Swiss's labor force participation. This coefficient provides direct evidence for the effect of different years of education on participation.

To improve the reliability of outcome in basic model, we need to construct a more comprehensive model that can incorporate key demographic and economic variables. The regression model with inclusions of the measured individual's age, nonlabor income, number of kids possessed (both aged below 7 and above 7) shows the relationship between education and labor force participation with controls. Controlled model:

participation_num = β_0 + β_1 (education) + β_2 (income) + β_3 (age) + β_4 (youngkids) + β_5 (oldkids) + ε



β_0 represents the intercept, where predicted probability of participation when all variables are zero. β_1 represents the predicted labor force change for each additional year of education. β_2 represents the percentage change in participation probability with one-unit increase in nonlabor income. β_3 represents the percentage change in participation probability with one-unit increase in age. β_4 represents the percentage change in participation probability with one-unit increase in possessed number of young kids. β_5 represents the percentage change in participation probability with one-unit increase in possessed number of older kids.

The significance and magnitude from β_2 to β_5 suggest nonlabor income, age, number of young and older kids are potential confounders. These confounders will at certain level impact the precision of true effect of different years of education. Adding these confounding variables in regression will serve to control confounders.

B. Sensitivity analysis

Variable	coefficient estimation	p-value	adjusted R-squared
education without control	-0.016843	0.00243 **	0.00938
education with control	-0.009871	0.0826 .	0.1046

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The basic model revealed an unexpected negative relationship between education and labor force participation. Without controlling for other factors, each additional year of education appeared to decrease the probability of individual Swiss's labor force participation by approximately 1.68 % (β_1 = -0.0168, $p < 0.01$). However, this model explained only about 1.05% of the variance in participation decisions (R^2 = 0.0105), suggesting that other important factors were being omitted.

The result of adding more control variables indicates a reversed education's role in labor force participation. In the controlled model, the education coefficient decreased substantially in both magnitude and statistical significance (β_1 = -0.0099, $p < 0.10$). While adding controls add more sensitivity to the model, the controlled model suggest that the negative relationship initially observed was almost untrue. The R-squared rised to 0.1097, indicating that these additional factors helped explain about 11% of the variance in participation decisions. Thus, years of education isn't a good factor to analyse its effect to participation.

After adding in more controls in the regression, the education coefficient decreased by 41.1% and the statistical significance changes from significant to insignificant, from -0.0168 ($p < 0.01$) to -0.0099 ($p < 0.10$). This dramatic reduction in both magnitude and statistical significance suggests that the initial strong negative relationship was untrue and driven by omitted variables. The weakening of both the coefficient size and its statistical significance indicates that education's negative effect on participation was primarily due to confounding variables rather than a true causal relationship.

In this case, we should turn our attention to analysing factors outside education. We can use the detected changes of different confounding variables' coefficients by putting education variable together with and without different factors. In controlled model, the coefficient of having how many young children for each individual presented to be the strongest influencing factor of reduced participation (β_4 = -0.2482, $p < 0.001$), suggesting that childcare responsibilities is a important factor to consider in labor market participation decisions. Non-labor income showed the second strongest effect (β_2 = -0.1892, $p < 0.001$), indicating that higher alternative income sources beside individual larbor earnings reduce the likelihood of labor force participation. Age also demonstrated a substantial negative relationship with participation (β_3 = -0.1224, $p < 0.001$), while the presence of older children had a negligible effect (β_5 = -0.0014, not significant).

The results of the sensitivity analysis suggest that the initial negative relationship between education and participation was likely driven by omitted variable bias. When adding the control variables, the education coefficient decreased substantially in both magnitude and statistical significance, indicating that other factors outside of education are more important determinants of participation decisions in Switzerland.

C. Omitted variable bias

There are several limitations in the analysis should be noted. Analysed in sensitivity analysis, the results of statistical insignificance in education coefficient and factors beside education show stronger impact than education on labor force participation indicate that there are other more important factors can influence labor force participation. Therefore, omitted variable bias may present.

From within the model, while linear terms are presented, nonlinear factors might influence the pattern in regression. Take age as an example. Since people at different age stage might have different capabilities to participate in labor force, the changes of participation due to age's effect probably doesn't have a linear pattern. Therefore, age squared as a omitted variable that has not presented in the model can be a way to measure the nonlinear relationship.

Age squared is likely positively related with education, because older people likely have already obtained more education attainment, while the relationship might be nonlinear due to the changing educational opportunities and norms over time. Age squared is likely negatively related with labor force participation, because labor force participation typically follows an inverted U-shape pattern with age, increasing fast in younger age and declining later. Therefore, the bias is negative when age squared is being omitted, meaning excluding age squared makes education's coefficient more negative.

Outside of the model, overall household income can be a omitted variable. Overall household income represents the synthetical income conditions revolving the individuals. Household income can be positively correlated with education, because there is more chance for people with higher educational backgrounds to have higher income for household; household income can be negatively correlated with labor force participation, because individuals with higher overall income from households probably have less need to participate in labor market through income effect. Therefore, the bias is negative when omitting household income from regression, making education's coefficient more negative than it should be. The model incorrectly attributes this income-driven reduction in participation to education.

Through our analysis that omitting age squared and overall household income will create negative bias, the counter-intuitive negative relationship between education and labor force participation may be reasonably explained.

IV. Conclusion

The investigation finds that education attainment likely does not have a direct impact on labor force participation in Switzerland. The relationship between education and participation appears to be more complex than a simple cause-and-effect dynamic. Other factors, such as family structure, economic conditions, and demographic characteristics, seem to be more important determinants of participation decisions.

The analysis revealed that when controlling for variables like non-labor income, age, and the presence of young children, the initially observed negative relationship between education and participation was largely explained away. This suggests the initial finding of education decreasing participation was driven by omitted variable bias rather than a true causal effect.

In fact, the data indicates that factors related to family responsibilities, such as having young children, and economic factors like non-labor income have stronger influences on labor force participation than educational attainment alone. This implies that policies aimed at increasing participation may be more effective if they focus on supporting working parents and addressing childcare constraints, rather than solely emphasizing educational initiatives.

Future research could build on these findings by incorporating additional variables, exploring non-linear relationships (e.g. the inverted U-shape pattern of age), and examining regional variations. Investigating how the education-participation link differs across demographic groups and economic contexts could also provide valuable insights. The use of panel data may also help control for unobserved individual characteristics that could impact both educational attainment and participation decisions.

In conclusion, this analysis highlights the complexity of labor market decisions and the importance of considering multiple, interacting factors when designing policies to influence participation rates. The results suggest that effective interventions should address practical barriers to employment, such as childcare access and age-related constraints, in addition to educational factors. By taking a more holistic view, policymakers may be better equipped to support labor force participation in Switzerland.

Appendix:

We first need to load data into R. Since the labor force participation here is a factor, it need to be converted to numeric to be counted in the regression.

```
# Load required packages
library(AER)
library(sandwich)
library(lmtest)


# Load Swiss Labor data
data("SwissLabor")

# Examine the structure of the data
str(SwissLabor)
# Convert participation to numeric
SwissLabor$participation_num <- as.numeric(SwissLabor$participation) - 1
```

```
## Create variables
education_years <- SwissLabor$education
nonlabor_income <- SwissLabor$income
labor_participation <- SwissLabor$participation
kid_number <- SwissLabor$youngkids+SwissLabor$oldkids
```

Constructing both single-variable regression model and multi-variable regression model will be able to compare on what extent will different variables influence the relationship between years of formal education and labor force participation

```
## Regression
# Basic regression (education only)
mod1 <- lm(participation_num ~ education, data=SwissLabor)
# Multiple regression (adding controls)
mod2 <- lm(participation_num ~ education + income + age + youngkids +oldkids, data=SwissLabor)
```

we need to visualize the relationships in both two models

```
## Visualize the relationship
#single variable regression
plot(SwissLabor$education, SwissLabor$participation_num,
     main = "Education vs Labor Force Participation",
     xlab = "Years of Education",
     ylab = "Labor Force Participation (0/1)",
     pch = 19,
     col = adjustcolor("blue", alpha=0.5))
abline(mod1, col="red", lwd=2)
#multiple variable regression
plot(SwissLabor$education+SwissLabor$income+SwissLabor$age+SwissLabor$youngkids+SwissLabor$oldkids, SwissLabor$participation_num,
     main = "Education vs Labor Force Participation",
     xlab = "Years of Education with controls",
     ylab = "Labor Force Participation (0/1)",
     pch = 19,
     col = adjustcolor("green", alpha=0.5))
abline(mod1, col="purple", lwd=2)
```

Using this grammar, the results for every factor included in the regression can be revealed

```r
## Sensitivity analysis
# Get robust standard errors for both models
robust_mod1 <- coeftest(mod1, vcov = vcovHC(mod1, type = "HC1"))
robust_mod2 <- coeftest(mod2, vcov = vcovHC(mod2, type = "HC1"))

# Print results
cat("\nBasic Model (with robust standard errors):\n")
print(robust_mod1)

cat("\nFull Model (with robust standard errors):\n")
print(robust_mod2)

# Compare models with full summary
cat("\nBasic Model Summary:\n")
print(summary(mod1))

cat("\nFull Model Summary:\n")
print(summary(mod2))
```